



GRADE 6

STEM OLYMPIAD

National Level Examination (NLE) 2026

SUBJECT CODE			
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Total Questions: 50

Total Marks: 50

Time: 75 Minutes

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED TO DO SO

- **Carefully Check Your Name, Class, Roll Number, And Other Required Details On The OMR Answer Sheet.**
- **Do Not Forget To Sign The OMR Sheet. Also Write Your Roll No. On The Question Paper Booklet.**
- **Use Only Black/Blue Ball Pen And HB Pencil To Fill The OMR Answer Sheet.**
- **All Questions Are Compulsory And Carry One Mark Each.**
- **Section A: Mathematics & Quantitative Reasoning (15 Questions); Section B: Science & Scientific Reasoning (15 Questions); Section C: Engineering, Technology & Data (10 Questions); Section D: STEM Olympiad Challenge (10 Questions).**
- **Only OMR Answer Sheets Will Be Evaluated.**
- **No Negative Marking For Incorrect Answers.**
- **You May Use The Space Provided On The Last Page For Rough Work (If Needed).**
- **Digital Devices Like Calculators, Smart Watches, Or Mobile Phones Are Not Allowed.**
- **After The Exam, Submit Both The OMR Sheet And The Question Paper To The Invigilator.**

SECTION A – MATHEMATICS & QUANTITATIVE REASONING

Q1. What is the value of $(-15) + (-8) - (-12) + 20$?

- A. 7
- B. 8
- C. 9
- D. 11

Q2. What is $\frac{3}{4} + \frac{5}{8}$?

- A. $\frac{7}{8}$
- B. $\frac{9}{8}$
- C. $\frac{11}{8}$
- D. $\frac{13}{8}$

Q3. What is 2.5×0.4 ?

- A. 0.1
- B. 1.0
- C. 10
- D. 100

Q4. The ratio 18:24 in simplest form is:

- A. 2:3
- B. 3:4
- C. 4:5
- D. 6:8

Q5. If $3x - 5 = 2x + 7$, what is the value of $2x + 1$?

- A. 12
- B. 24
- C. 25
- D. 26

Q6. A rectangle has a perimeter of 38cm. If its length is 12 cm, What is the area of the rectangle?

- A. 84 cm^2
- B. 96 cm^2
- C. 60 cm^2
- D. 42 cm^2

Q7. A cuboid measures 5 cm x 4 cm x 3 cm. What is its volume?

- A. 12 cm^3
- B. 20 cm^3
- C. 47 cm^3
- D. 60 cm^3

Q8. The mean of 6, 8, 10, 12 and 14 is:

- A. 9
- B. 10
- C. 11
- D. 12

Q9. Which number is prime?

- A. 21
- B. 29
- C. 33
- D. 39

Q10. The LCM of 8 and 12 is:

- A. 16
- B. 20
- C. 24
- D. 48

Q11. If 5 notebooks cost Rs. 210, what do 8 notebooks cost at the same rate?

- A. Rs 288
- B. Rs 320
- C. Rs 336
- D. Rs 360

Q12. A line segment has:

- A. one endpoint
- B. two endpoints
- C. no endpoints
- D. three endpoints

Q13. An angle of 270° is:

- A. acute
- B. obtuse
- C. reflex
- D. straight

Q14. A car travels 180 km in 3 h at constant speed. Its speed is:

- A. 50 km/h
- B. 60 km/h
- C. 70 km/h
- D. 90 km/h

Q15. What is 25% of 240?

- A. 40
- B. 50
- C. 60
- D. 80

SECTION B – SCIENCE & SCIENTIFIC REASONING

Q16. Which nutrient is mainly needed for body growth and repair?

- A. Protein
- B. Water only
- C. Roughage only
- D. Mineral salt only

Q18. Which method separates an insoluble solid from a liquid?

- A. Filtration
- B. Evaporation only
- C. Magnetism only
- D. Freezing

Q17. Which plant tissue carries water upward from roots?

- A. Xylem
- B. Phloem
- C. Petal
- D. Stomata

Q19. Which joint allows movement in many directions, as in the shoulder?

- A. Hinge
- B. Ball-and-socket
- C. Fixed
- D. Pivot only

Q20. Which change is chemical?

- A. Melting wax
- B. Dissolving sugar
- C. Burning paper
- D. Freezing water

Q21. The standard unit of temperature is ?

- A. Fahrenheit
- B. kelvin
- C. °Celsius
- D. Joule

Q22. A shadow is shortest around midday mainly because the Sun is:

- A. higher in the sky
- B. below the horizon
- C. blocked by Earth
- D. producing less light

Q23. A simple electric circuit works when it provides:

- A. a complete conducting path
- B. only one wire touching the bulb
- C. an open switch
- D. no cell

Q24. Which end of a freely suspended magnet points towards the geographical North pole?

- A. North-seeking pole
- B. South-seeking pole
- C. Magnetism heats it
- D. The spoon creates heat

Q25. Why does a metal spoon in hot tea become warm?

- A. Heat is conducted through the metal
- B. Sound moves through it
- C. Magnetism heats it
- D. The spoon creates heat

Q26. Which gas is most abundant in air?

- A. Oxygen
- B. Nitrogen
- C. Carbon dioxide
- D. Water vapour

Q27. Which process returns water vapour to liquid droplets in clouds?

- A. Condensation
- B. Evaporation
- C. Sublimation
- D. Melting

Q28. Which adaptation best helps a fish move through water?

- A. Streamlined body
- B. Broad flat feet
- C. Dry scales only
- D. Long eyelashes

Q29. Four different waste items were buried in moist soil under identical conditions. After six weeks, which items will show the Highest rate of decomposition without leaving residue?

- A. A laminated paper
- B. Fruits peels and leftover cooked rice
- C. A thin, single-use plastic bag
- D. An aluminum soft drink can

Q30. Which force slows a sliding book on a table?

- A. Friction
- B. Magnetism only
- C. Light
- D. Sound

SECTION C – ENGINEERING, TECHNOLOGY & DATA

Q31. A bridge truss uses triangles because triangles are:

- A. rigid and resist shape change
- B. always lighter than air
- C. magnetic
- D. transparent

Q32. A student compares evaporation from two dishes. Which condition should differ in a fair test of surface area?

- A. Only the exposed surface area
- B. Temperature of water
- C. Amount of water
- D. Sunlight exposure

Q33. A lever has the load 20 cm from the fulcrum and effort 60 cm from the fulcrum. Ignoring losses, the effort needed is:

- A. greater than the load
- B. less than load
- C. One-third of the load
- D. zero

Q34. Which circuit arrangement keeps other bulbs glowing if one bulb fails?

- A. Parallel
- B. Series only
- C. Open circuit
- D. No wires

Q36. A sensor records 20°C, 22°C, 24°C, 26°C. What is the rate of change per reading?

- A. 1°C
- B. 2°C
- C. 3°C
- D. 4°C

Q37. A robot moves 5 m East, 3 m North, 2 m West. Its final position from the start is:

- A. 3 m East and 3 m North
- B. 7 m East and 3 m North
- C. 3 m West and 3 m North
- D. 5 m East and 1 m North

Q38. A paper airplane travels farther after its wings are adjusted. What should be done to test whether the wing change caused the improvement?

- A. Repeat trials while keeping throw method the same
- B. Use a different thrower each time
- C. Change wing shape and throw strength together
- D. Use only one flight

Q39. A 500 W device runs for 2 h. How much energy does it use in kilowatt-hours?

- A. 0.5 kWh
- B. 1 kWh
- C. 2 kWh
- D. 1000 kWh

Q35. A 2 kg object has greater mass than a 500 g object by:

- A. 0.5 kg
- B. 1.0 kg
- C. 1.5 kg
- D. 2.5 kg

Q40. A water tank holds 150 L. It is $\frac{3}{5}$ full. How much water is in it?

- A. 60 L
- B. 75 L
- C. 90 L
- D. 120 L

SECTION D – STEM OLYMPIAD CHALLENGE

Q41. If a 12-tooth gear drives a 36-tooth gear, the larger gear makes one turn when the smaller makes:

- A. 1 turn
- B. 2 turns
- C. 3 turns
- D. 6 turns

Q42. Which graph is best for showing how temperature changes over time?

- A. Line graph
- B. Pie chart only
- C. Pictograph only
- D. Venn diagram

Q43. A solar cell changes light energy mainly into:

- A. electrical energy
- B. sound energy
- C. chemical energy only
- D. gravitational energy

Q44. Which design most reduces heat transfer from a hot bottle?

- A. Vacuum layer with insulating stopper
- B. Bare metal wall
- C. Wide open top
- D. Thin copper covering

Q45. A machine lifts a 100 N load using 50 N effort. Its mechanical advantage is:

- A. 0.5
- B. 1
- C. 2
- D. 50

Q46. A seed experiment uses 20 seeds in light and 20 in dark, both with equal water and temperature. What is the independent variable?

- A. Light condition
- B. Number of seeds
- C. Water amount
- D. Seed type

Q47. A cyclist covers 6 km in 20 min. At the same speed, how far in 1 h?

- A. 12 km
- B. 15 km
- C. 18 km
- D. 24 km

Q48. Which observation most strongly suggests a new substance formed?

- A. Gas forms and colour changes during a reaction
- B. Ice melts
- C. Salt dissolves
- D. Water freezes

Q49. A 10 m ramp rises 2 m vertically. Compared with lifting straight up, the ramp generally requires:

- A. less force over more distance
- B. more force over less distance
- C. no force
- D. the same force and no extra distance

Q50. A STEM team tests three tower designs. The best evidence for choosing one is:

- A. repeatable load-test data
- B. the tallest student's opinion
- C. the prettiest colour
- D. one lucky trial